# The Arrow of Time Is Irreversible Computation

**Gary Abraham Bernstein** Independent Researcher ORCID: 0009-0009-1761-2867

## Abstract

Computation is deterministic forward: 1 + 1 = 2. It is underdetermined backward: 2 = x + y has infinitely many solutions. The inverse function does not exist. This is a fact about mathematics, true in every possible world.

A system running backward cannot compute, model, observe, or self-model, let alone sustain consciousness. Every operation has many possible inputs; prediction requires computing consequences from causes, and causes are underdetermined from consequences; modeling requires updating representations, and updates are many-to-one operations with no inverse. No observer can exist or experience in such a direction. Retrodiction, where it succeeds, is a new forward computation over constrained hypotheses, not inversion. Reversible computing does not evade the point: reversible machines compute a different bijection and cease to model once no compression, overwrite, or reset is allowed.

In a block universe where all states exist, backward states are present but nothing can compute through them. This is the arrow of computation. It is the structural foundation for the time asymmetry that algorithmic probability quantifies. Forward computation does not increase Kolmogorov complexity relative to the generating rule: given simple laws and simple initial conditions, any later state is compactly described by "apply this rule for t steps." Running backward requires specifying which of infinitely many possible inputs produced each output, an arbitrarily high specification cost. The asymmetry is not in the end state but in the backward specification of the path. The computational arrow stands without algorithmic probability; algorithmic probability's time arrow is one way of measuring the asymmetry that irreversible computation creates.

Deutsch argues the arrow is perspectival because the global wavefunction evolves unitarily. His own constructor theory (Deutsch, 2013) contradicts him: it defines physics through local operations, and those operations are many-to-one. The arrow lives at the level his own theory selects. This affirms Many-Worlds: MWI is the only interpretation where the substrate stays reversible while the arrow exists at the function level. Collapse breaks substrate reversibility when the arrow already exists in the function, adding specification cost for no gain. Superdeterminism is compatible for the same reason: deterministic initial conditions, forward arrow, no collapse.

Consciousness is a further consequence: sentience requires modeling (Bernstein, 2026f), and modeling requires forward computation.

**Keywords:** computation, arrow of time, irreversibility, block universe, algorithmic information

*One plus one equals two. In reverse, two has infinite random solutions. Time backwards makes conscious observation impossible.*

## 1. The Problem

In a four-dimensional block universe, all spacetime points exist equally. No moment objectively passes. If so, why do conscious systems remember one temporal direction and anticipate the other? Why do records point one way? Why do explanations run from prior to later rather than later to prior?

The usual answer invokes entropy. A low-entropy boundary condition makes one temporal direction distinctive, and systems embedded in such a world form memories along that gradient. That answer may be correct as far as it goes, but it begins too late. Before entropy, before memory, before consciousness, there is a more basic question: why does computation itself have a direction?

The claim of this paper is simple. The relevant asymmetry is logical before it is thermodynamic. Many tasks central to modeling, abstraction, memory, and task completion are many-to-one. Such maps have a deterministic forward direction but no inverse function. Once that is recognized, the directionality of prediction, modeling, memory, and construction follows.

## 2. Core Theorem

Let $f$: $X \rightarrow Y$. If $f$ is many-to-one, then there is no function $f^{\ 1}$: $Y \rightarrow X$ that recovers a unique input for every output. The reverse relation exists, but it is not a function. A present output may correspond to many possible past inputs. Any "inverse" must return a set of preimages or a probability distribution over them, not a single input.

Addition illustrates the pattern. The preimages of 2 include (0,2), (1,1), (−1,3), and infinitely many others. Multiplication shows the same shape: $12 = \mathrm{x} \times \mathrm{y}$ has preimages (1,12), (2,6), (3,4), and so on. The reverse problem selects among compatible antecedents. It is a different problem from the same computation run backward.

The functions that matter for minds and for macroscopic physical tasks are the ones relevant to modeling, coarse-graining, classification, thresholding, abstraction, updating, and resettable task performance. Microscopic laws need not themselves be many-to-one for the claim to hold.

The thesis can be stated precisely:

> **Theorem (Computational Arrow).** No computation that models by compression can run backward as the inverse of the same function.

Equivalently, once a map has been used to merge many inputs into one compressed state, there is no single-valued map from that state back to a unique input. Any “inverse” must return a set of preimages or a probability distribution over them.

## 3. Modeling and Compression

A model is not a bare mirror of the world. It is a selective internal representation. That means compression. Multiple detailed world-states must map to the same representation if the representation is to generalize, ignore noise, track invariants, and guide action.

A visual system maps many retinal microstates to one percept. A neuron fires when summed synaptic inputs cross threshold, so thousands of input combinations produce the same spike. A concept maps many token experiences to one category. A memory system maps a detailed prior episode to an encoded trace. A scientific model maps many details to a tractable state-space description. In each case, the model is useful because it throws detail away.

The same logic cascades through pattern recognition, observation, and self-modeling. A pattern classifier compresses many signals into one label. Observation compresses physical state into record. Self-modeling compresses prior states into current ones. Each is forward-only for the same reason addition is: the generating function is many-to-one. On a self-modeling account of sentience (Bernstein, 2026f), feeling and consciousness inherit the arrow through this same structure.

Once many possible antecedents map to one representational state, the inverse is underdetermined. The present representation does not uniquely contain the path by which it was formed. There is no function from the representational state alone back to its unique generating history.

This is also why memory points one way. A memory state is produced from prior interaction; it is not produced from later states. The current state encodes information about prior states because it was computed from them; it does not encode future states in the same way because it was not computed from them.

## 4. Retrodiction Is Not Inversion

A natural objection is immediate: science retrodicts all the time. Cosmology reconstructs the early universe. Paleontology reconstructs extinct lineages. Forensics reconstructs prior events from traces. In what sense can backward computation be impossible?

Retrodiction is a new forward computation over hypotheses. It does not invert the original map.

Once $f$ is many-to-one, the reverse task is not $f^{-1}(y)$, because $f^{-1}$ is not a function. What one can compute instead is a distribution $P(X \mid Y, C)$, where $C$ contains laws, priors, boundary conditions, auxiliary records, and background knowledge. That is a forward inferential procedure applied to present data plus constraints.

It narrows a space of compatible antecedents. It does not uniquely recover “the” input from the output alone.

This identifies where the asymmetry lies. Recovering the past always requires added structure beyond the output itself. The reverse problem is underdetermined unless supplied with constraints not contained in the target state alone.

Retrodiction therefore presupposes the computational arrow rather than overturning it.

## 5. Reversible Computation

The strongest technical objection comes from Bennett-style reversible computation. If a machine can be engineered so that each step is bijective, then why say computation is intrinsically directional?

Because a reversible machine computes a different function.

If ordinary addition is treated as (3,4)   7, the inverse does not exist as a function. A reversible implementation does not compute the inverse of that map. It computes something like (3,4,0)   (3,4,7), preserving the inputs alongside the output so the whole transformation remains bijective. That is an enlarged embedding of the original into a different reversible map, not the original computation run backward.

The claim here concerns task-level computations used for modeling and cognition: they are irreducibly lossy. Once a system classifies, abstracts, overwrites, thresholds, or returns to a ready state, it has performed a many-to-one task.

A mind that never discards, compresses, or overwrites anything ceases to function as a modeler. Learning requires replacement of one representational state by another. Memory requires selective retention. Concepts require abstraction. Action-guiding systems require resettable readiness. Those are precisely the points at which task-level irreversibility reappears.

Reversible computing reminds us that substrate-level reversibility and task-level irreversibility are different notions. It is not a counterexample.

The data-processing inequality is the information-theoretic form of this fact: no processing of an output can increase mutual information with the source (Cover & Thomas, 2006). Once information has been discarded by a many-to-one map, no post-processing of the output can reconstruct the missing distinctions without auxiliary information. The present argument generalizes that logic to modeling tasks.

## 6. Unitarity and Task-Level Irreversibility

Another objection comes from quantum theory. If the universal wavefunction evolves unitarily, then the deepest dynamics are one-to-one. Why not conclude that the arrow is merely perspectival?

The present argument concerns what kinds of functions are instantiated by observers, records, and physical tasks inside the substrate. It is a separate question from whether the deepest substrate is reversible.

A globally unitary substrate can still contain subsystems whose effective or task-level operations are many-to-one. Decoherence, coarse-graining, branch-relative records, and macroscopic state descriptions all trade on such reductions. The arrow defended here lives there: representation, construction, and memory use forward-only task structure even when embedded in a reversible whole.

The central distinction is:

- **Substrate-level reversibility:** the total dynamics may be bijective.
- **Task-level irreversibility:** the operations that produce records, models, measurements, classifications, and ready-state constructors are many-to-one.

The latter is enough for the arrow of experienced time. A reversible whole does not abolish local asymmetry any more than a reversible library abolishes the asymmetry between writing a summary and reconstructing every omitted sentence from that summary alone.

## 7. Deutsch and Constructor Theory

Deutsch's strongest line treats the arrow as perspectival. On his view, the universal wavefunction evolves unitarily and reversibly; apparent irreversibility arises from decoherence, branch-relative records, and the information-processing perspective of embedded observers. The present argument agrees that the substrate can be unitary and no stochastic collapse is needed.

But Deutsch's own constructor theory defeats the global framing. Constructor theory defines physics through local tasks: what constructors can reliably perform on substrates and return to a ready state. Any task that returns a constructor to its ready state is many-to-one over histories. Multiple execution histories map to the same ready-state description. The arrow lives at exactly the level his own method selects.

So Deutsch is right about global reversibility and wrong that it removes the arrow. A further K-based argument reinforces the point: a universe described forward from initial conditions I has total complexity K(L) + K(I ), both low. Described backward from a final state, K is enormous. The arrow follows from the K-decomposition alone, with no reference to the global wavefunction. The global wavefunction adds specification cost without adding explanatory power for the arrow. MWI adds explanatory power for the absence of collapse, not for the direction of time. Once physics is described through tasks (measurement, preparation, discrimination, memory, stabilization, resettable action), irreversibility reappears at the level that matters for agents.

The right conclusion is more precise:

**Global reversibility is compatible with a real computational arrow at the level of tasks.**

That point favors Everettian pictures over collapse pictures, because collapse adds substrate-level irreversibility where task-level irreversibility already suffices. It also leaves room for superdeterministic pictures, provided they preserve the same forward-only task structure. The central thesis does not depend on choosing between those options.

## 8. Thermodynamics

How does this relate to entropy?

Thermodynamic irreversibility concerns physical implementation: dissipative processes, record formation, erasure costs, and low-entropy boundary conditions. The computational arrow defended here concerns logical structure: whether a task defines a function in one direction and only a relation in the other.

These are different. Thermodynamics explains why one temporal direction in our universe is associated with increasing entropy and robust records. The computational argument explains why record-like systems, once they exist at all, operate from input to output rather than output to unique input.

The most plausible relation: thermodynamics implements at scale what computation already requires in logic. Minds are physical systems; their forward-only computational structure is realized through thermodynamic processes. The arrows are aligned without being identical.

## 9. Algorithmic Probability and Complexity

A confusion worth avoiding is the idea that forward evolution somehow makes later states intrinsically high in Kolmogorov complexity. Given simple laws and simple initial conditions, a later state remains compactly describable by "apply this rule for $t$ steps." The real cost lies in specifying which compatible history led there without using the rule.

There are three distinct levels in play:

1. **Function-level irreversibility:** many-to-one maps lack inverse functions.
2. **State-description complexity:** a state may be simple or complex relative to a coding and a generating rule.
3. **History-specification complexity:** selecting which compatible preimage or history led to a present outcome generally requires added specification.

The present paper concerns the first level. Algorithmic probability, where relevant, lives mostly at the third. If many histories are compatible with a present state, then singling out one history or one branch involves additional descriptive cost unless a short rule picks it out. Algorithmic-probability asymmetries can be seen as ways of measuring the burden imposed by underdetermined inverse reconstruction.

Algorithmic probability is therefore not a premise of the computational arrow. At most, it is one way of quantifying a downstream consequence.

## 10. Conclusion

In the block universe, every temporal neighbor exists, yet the present state inherits information only from the states that generated it. Backward states exist; the inverse computation through the same task does not. Many-to-one computations do not admit inverse functions. Modeling, memory, prediction, abstraction, and construction rely on many-to-one tasks. No system that models by compression can run backward through those tasks as inverses of themselves. Retrodiction, where possible, is achieved by new forward computations over constrained hypothesis spaces.

This yields a computational arrow of time. It accommodates global reversibility. It needs no collapse. It preserves retrodiction. It aligns with thermodynamic irreversibility without depending on any particular thermodynamic theory, and can be measured in part through algorithmic-probability considerations.

Sentience inherits this arrow through the modeling requirement (Bernstein, 2026f): any system that models itself is subject to the same forward-only task structure.

One plus one leads to two. Two does not lead back to one plus one. The arrow is that inverse reconstruction is not, in general, a function. Later states need not be intrinsically more complex.

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